

OITA, Japan

WMO: 478150

Lat: 33.2362N

Lon: 131.6200E

Elev: 13

StdP: 101.16

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.3	0.7	-8.7	1.8	3.7	-7.5	2.0	4.4	7.6	7.2	7.1	6.8	2.9	180	0.369

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.2	33.6	25.4	32.6	25.3	31.6	25.0	26.6	31.2	26.0	30.6	25.6	29.9	3.8	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.3	20.4	28.8	24.8	19.8	28.4	24.3	19.3	28.1	83.1	31.4	80.8	30.7	78.7	30.0	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.1	6.2	5.4	-2.2	35.6	1.0	0.9	-2.9	36.2	-3.5	36.7	-4.1	37.2	-4.9	37.9		
			-3.9	27.3	1.2	0.7	-4.8	27.7	-5.5	28.1	-6.1	28.5	-7.0	29.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.0	6.4	7.2	10.2	14.9	19.5	22.8	27.2	28.2	24.6	19.4	13.8	8.8	(6)
(7)		DBStd	7.89	2.35	2.80	2.91	2.85	2.19	2.32	2.14	1.74	2.48	2.65	2.99	2.79	(7)
(8)		HDD10.0	296	113	87	33	2	0	0	0	0	0	3	58		(8)
(9)		HDD18.3	1506	369	312	253	107	13	0	0	0	18	138	297		(9)
(10)		CDD10.0	2842	3	9	39	149	295	384	534	564	437	291	118	19	(10)
(11)		CDD18.3	1009	0	0	0	4	49	135	276	306	187	51	2	0	(11)
(12)		CDH23.3	8589	0	0	2	16	183	693	2781	3401	1347	166	2	0	(12)
(13)		CDH26.7	2892	0	0	0	1	14	142	1029	1356	335	14	0	0	(13)
(14)	Wind	WSAvg	2.6	2.9	2.8	2.8	2.6	2.5	2.1	2.3	2.5	2.5	2.5	2.6	2.9	(14)
(15)	Precipitation	PrecAvg	1689	47	64	98	128	146	288	242	168	241	141	70	39	(15)
(16)		PrecMax	2863	182	175	322	246	359	631	702	731	701	621	209	148	(16)
(17)		PrecMin	1075	1	8	26	17	20	42	37	2	5	15	5	0	(17)
(18)		PrecStd	354	40	40	51	48	75	135	152	141	165	131	50	34	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.7	18.9	21.4	25.2	28.4	32.1	34.6	35.2	33.0	28.4	23.1	18.2	(19)
(20)		0.4%	MCWB	10.9	14.5	14.3	17.2	18.8	24.4	25.4	25.8	24.9	22.4	17.6	13.4	(20)
(21)		2%	DB	13.6	15.5	18.8	23.1	26.6	29.7	33.4	34.0	31.3	26.4	21.4	16.4	(21)
(22)		2%	MCWB	9.4	11.4	13.3	15.7	18.1	23.0	25.4	25.6	24.3	20.9	16.8	12.1	(22)
(23)		5%	DB	12.2	13.6	17.0	21.4	25.4	28.3	32.5	33.0	29.8	25.1	20.2	15.0	(23)
(24)		5%	MCWB	8.3	9.8	11.8	15.0	17.7	22.2	25.2	25.4	23.6	19.9	16.0	11.0	(24)
(25)		10%	DB	11.0	12.1	15.5	20.0	24.1	26.8	31.3	32.0	28.6	24.0	19.0	13.8	(25)
(26)		10%	MCWB	7.3	8.3	11.1	14.3	17.4	21.3	25.0	25.2	23.1	19.1	15.1	10.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.0	15.2	16.2	19.7	22.2	25.2	27.1	27.4	26.2	24.0	19.7	15.3	(27)
(28)		0.4%	MCDWB	14.8	18.5	19.0	21.5	24.9	29.7	31.8	32.6	30.1	26.6	21.7	16.7	(28)
(29)		2%	WB	10.6	12.6	14.6	18.0	20.8	24.3	26.4	26.7	25.3	22.8	18.3	13.3	(29)
(30)		2%	MCDWB	12.3	14.9	17.4	20.8	23.3	28.1	31.1	31.6	28.9	25.3	20.1	15.2	(30)
(31)		5%	WB	9.0	10.7	13.1	16.8	19.8	23.6	25.9	26.2	24.7	21.3	17.1	11.9	(31)
(32)		5%	MCDWB	11.4	12.7	15.9	20.0	22.7	26.9	30.5	30.9	28.1	23.9	19.2	14.2	(32)
(33)		10%	WB	7.7	9.1	11.7	15.6	19.0	22.8	25.4	25.8	24.1	20.0	15.8	10.6	(33)
(34)		10%	MCDWB	10.5	11.6	14.6	18.8	22.2	25.7	29.8	30.2	27.5	22.7	18.4	13.3	(34)
(35)	Mean Daily Temperature Range		MDBR	7.5	7.6	8.1	8.5	8.3	6.5	6.7	7.2	6.6	7.4	7.8	7.7	(35)
(36)		5% DB	MCDBR	9.5	9.8	10.6	11.0	10.5	8.5	8.4	8.6	8.0	8.3	8.7	9.3	(36)
(37)			MCWBR	6.4	6.9	6.8	6.1	5.2	3.6	3.0	3.0	3.3	4.5	5.6	6.2	(37)
(38)			MCDBR	8.1	8.5	9.0	8.3	7.6	6.5	7.5	7.8	6.8	6.1	6.8	7.6	(38)
(39)		5% WB	MCWBR	6.6	6.9	6.8	5.5	4.8	3.1	3.0	3.1	3.1	3.9	5.0	6.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.419	0.462	0.525	0.549	0.557	0.567	0.565	0.538	0.508	0.464	0.428	0.410	(40)	
(41)		taud	2.101	2.018	1.868	1.865	1.910	1.966	2.014	2.082	2.125	2.161	2.175	2.145	(41)	
(42)		Ebn at Noon	764	771	754	758	757	746	745	759	762	764	752	748	(42)	
(43)		Edh at Noon	128	154	193	202	195	184	175	161	148	132	118	117	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.43	3.06	4.09	4.82	5.13	4.27	4.91	5.08	4.08	3.43	2.63	2.20	(44)	
(45)		RadStd	0.21	0.27	0.27	0.47	0.58	0.57	0.68	0.50	0.46	0.37	0.25	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+51
(47)	Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page